



Spontaneous regression of tumors post-chemotherapy in a patient with relapsed small cell lung cancer: a case report

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Background: Spontaneous regression (SR) of small cell lung cancer (SCLC) is an extremely rare phenomenon, and its underlying mechanisms remain unclear. Previous reports have suggested that immune-related factors, including paraneoplastic neurological syndrome (PNS) and anti-Hu antibodies, may be involved. In this report, we present a case of SCLC that demonstrated SR after initial progression, and explore the potential immunological mechanisms, particularly focusing on changes in the tumor microenvironment.

Case Description: A 78-year-old Japanese male with a significant smoking history presented with two pulmonary nodules. Surgical resection revealed non-small cell lung cancer (NSCLC) and SCLC (stage IA2). Seven months later, SCLC recurrence was confirmed via transbronchial needle aspiration (TBNA) of the subcarinal lymph nodes. First-line chemotherapy with carboplatin and etoposide resulted in a good response; however, disease progression was observed later. Second-line treatment with carboplatin and nanoparticle albumin-bound (nab)-paclitaxel was initiated, but therapy was then discontinued because of declining performance status. Unexpectedly, without further treatment, the tumor and serum gastrin-releasing peptide precursor (pro-GRP) levels began to decrease over 6 months, indicating SR. Immunohistochemical analysis revealed an increase in cluster of differentiation (CD) 8-positive T cells and a decrease in regulatory T cells in the recurrent tumor, thus suggesting immune modulation.

Conclusions: This case of SCLC showed SR following chemotherapy, which was likely influenced by immune-related mechanisms. Changes in the tumor microenvironment, particularly an increase in cytotoxic T cells and a reduction in regulatory T cells, may contribute to SR. Nab-paclitaxel may contribute to the modulation of the tumor immune microenvironment. Therefore, understanding these mechanisms could provide insights into tumor immunity and potential therapeutic strategies.

Keywords: Small cell lung cancer (SCLC); spontaneous regression (SR); CD8-positive T cells; nab-paclitaxel; case report

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Introduction

Small cell lung cancer (SCLC) accounts for approximately 15% of all lung cancers and is characterized by rapid proliferation, early metastasis and poor prognosis (1). In contrast, non-small cell lung cancer (NSCLC) comprises the remaining 85%, and the treatment landscape has evolved with the advent of targeted therapies and immune checkpoint inhibitors (ICIs) (2). Although most patients with SCLC initially respond to chemotherapy, recurrence is common and long-term survival remains poor. ICIs such as atezolizumab or durvalumab have been incorporated into first-line therapy for extensive-stage SCLC, modestly improving overall survival (3).

Spontaneous regression (SR) of SCLC is very rare, and the mechanisms of SR in cancer are unclear. Although previous reports have suggested that immune-related factors may be involved, SR occurred without paraneoplastic neurological syndrome (PNS) or detectable immune-related syndromes in this case.

Here, we describe a case of SCLC that spontaneously regressed and discuss the mechanisms of SR based on

immunohistochemistry results and previous literature. We present this article in accordance with the CARE reporting checklist (available at <https://tlcr.amegroups.com/article/view/10.21037/tlcr-2025-578/rc>).

Case presentation

A 78-year-old Japanese male visited our hospital with nodules in the left S⁶ and S⁹ on chest computed tomography (CT) in May 2021. The patient was a 58-pack per year smoker with a history of arteriosclerosis obliterans, aortic stenosis, hypertension, and diabetes mellitus. He underwent bronchoscopy, but a diagnosis could not be made. Left lower lobectomy with mediastinal lymph node dissection was performed in July 2021. The left S⁶ nodule was diagnosed with pT1bN0M0 stage IA2 NSCLC and the left S⁹ nodule was diagnosed with pT1bN0M0 stage IA2 SCLC.

Seven months later, recurrence of lung cancer was suspected on chest CT, and TBNA of the subcarinal lymph nodes revealed recurrence of SCLC. The patient had no symptoms or abnormalities on physical examination. The Eastern Cooperative Oncology Group (ECOG) performance status (PS) was 0. Laboratory examination revealed a gastrin-releasing peptide precursor (pro-GRP) level of 406 pg/mL (normal range, <46.0 pg/mL). Chest CT revealed a mass adjacent to the previous lobectomy site and enlargement of the subcarinal lymph nodes, which showed emphysematous and interstitial pneumonia (*Figure 1*).

In April 2022, the patient was treated with four cycles of chemotherapy based on carboplatin [area under the curve (AUC) 5, on Day 1] and etoposide (100 mg/m² on days 1, 2, and 3) as first-line treatment, in accordance with the standard treatment for SCLC with interstitial pneumonia, with a good response. However, 6 months later, mediastinal lymph node enlargement was observed on CT, and the patient was diagnosed with progressive disease.

In December 2022, the patient was treated with carboplatin (AUC 5, on day 1) and nab-paclitaxel (80 mg/m² on days 1, 8, and 15) as second-line treatment. In the situation of SCLC with interstitial pneumonia, where there is no standard treatment for second-line therapy or beyond, combination chemotherapy with carboplatin and nab-paclitaxel was selected because it seems to be safe in patients with interstitial pneumonia (4). After four cycles of chemotherapy, the ECOG PS showed a declining trend, leading to the cessation of treatment.

During the course of treatment, the subcarinal lymph nodes, recurrent mass, and the pro-GRP level reflected

Highlight box

Key findings

- This case describes a rare instance of spontaneous regression (SR) in small cell lung cancer (SCLC) following the cessation of second-line chemotherapy. Immunohistochemical analysis revealed an increase in CD8-positive T cells and a decrease in regulatory T cells, thus suggesting immune system involvement in the tumor regression.

What is known and what is new?

- SR of SCLC is extremely rare, and when reported, it is often associated with paraneoplastic syndromes or immune responses.
- In this case, SR occurred without paraneoplastic neurological syndrome (PNS) or any detectable immune-related syndromes. The immunohistochemical findings suggest that prior chemotherapy, particularly with nab-paclitaxel, may have modulated the tumor microenvironment to favor immune-mediated regression.

What is the implication, and what should change now?

- This case may suggest a possible involvement of immune modulation in the SR of SCLC. Although the exact mechanisms remain unclear, the observation presented raises the question of how chemotherapy might influence the tumor immune microenvironment, particularly the balance between cytotoxic and regulatory T cells. Further investigation into tumor immunity in SCLC could therefore provide insights into novel therapeutic approaches.

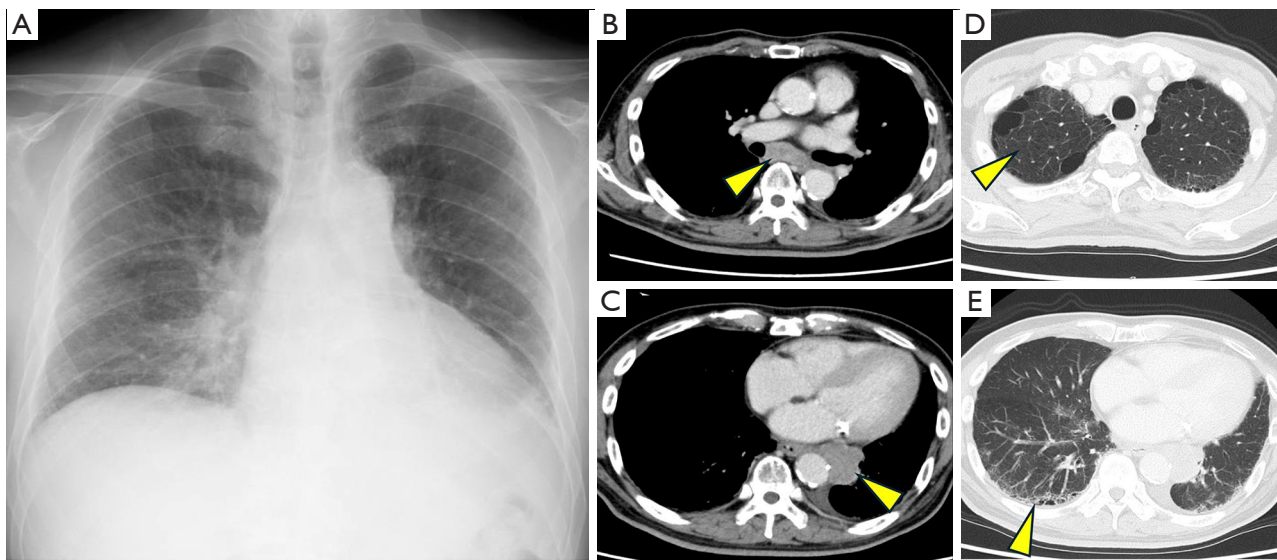


Figure 1 Chest X-ray film and CT the present case. (A) No special findings in April 2022. Chest CT scan revealed enlargement of subcarinal lymph node (B) and a mass adjacent to the previous lobectomy (C), and the background lungs showed emphysematous (D) and interstitial pneumonia (E). Each arrowhead points to the site of disease. CT, computed tomography.

the disease status (*Figure 2*), highlighting the clinical significance of pro-GRP as a tumor marker in this case.

It was expected that the tumors would increase in size and the pro-GRP level would increase post-treatment. However, after initial enlargement, the tumors reduced in size, and the pro-GRP level showed a downward trend. In this case, because the tumor reduced in size for 6 months without treatment, it was considered to be SR of SCLC.

All of the procedures performed in this case report were in accordance with the ethical standards of Juntendo University Hospital and with the Declaration of Helsinki and its subsequent amendments. Written informed consent for publication of this case report and accompanying images was not obtained from the patient or the relatives after all possible attempts were made.

Discussion

SR is defined as the partial or complete disappearance of a malignant tumor in the absence of all treatment, or in the presence of therapy that is considered inadequate to exert a significant influence on neoplastic diseases. The frequency of SR is reported to be one in 60,000 to 100,000 people (5).

Of the 40 cases of SR of lung cancer reported between 2006 and 2011, SCLC accounted for 20% of the histopathological types. There was a tendency for SCLC

to be more common than the general frequency, based on histopathology.

Referring to case reports on SR of SCLC, most case reports of SCLC suggest immunological involvement, such as PNS and positivity for anti-Hu antibodies (6). Given that previous reports have suggested its involvement, we conducted serological testing for PNS in our case; however, the results were negative, and no neurological symptoms were observed. There have also been cases where it is thought that immune system activation by an invasive procedure, such as a biopsy or bronchoscopy, leads to regression of tumors.

To investigate the mechanism of SR in the current case, we performed additional staining, focusing on the tumor microenvironment. The patient had a recurrence and subsequently received carboplatin, etoposide, and nab-paclitaxel therapy. Post cessation of chemotherapy, the tumor showed a tendency to increase in size, but then spontaneously regressed for 6 months. Based on the immunohistochemistry and medical history, we considered the mechanism of the SR. An increase in CD8-positive T cells and a decrease in regulatory T cells were observed in the TBNA samples obtained at the time of recurrence compared to those in the surgical specimens. CD8-positive T cells were quantified and showed an increase (*Figure 3*). Regulatory T cells were determined to decrease visually,

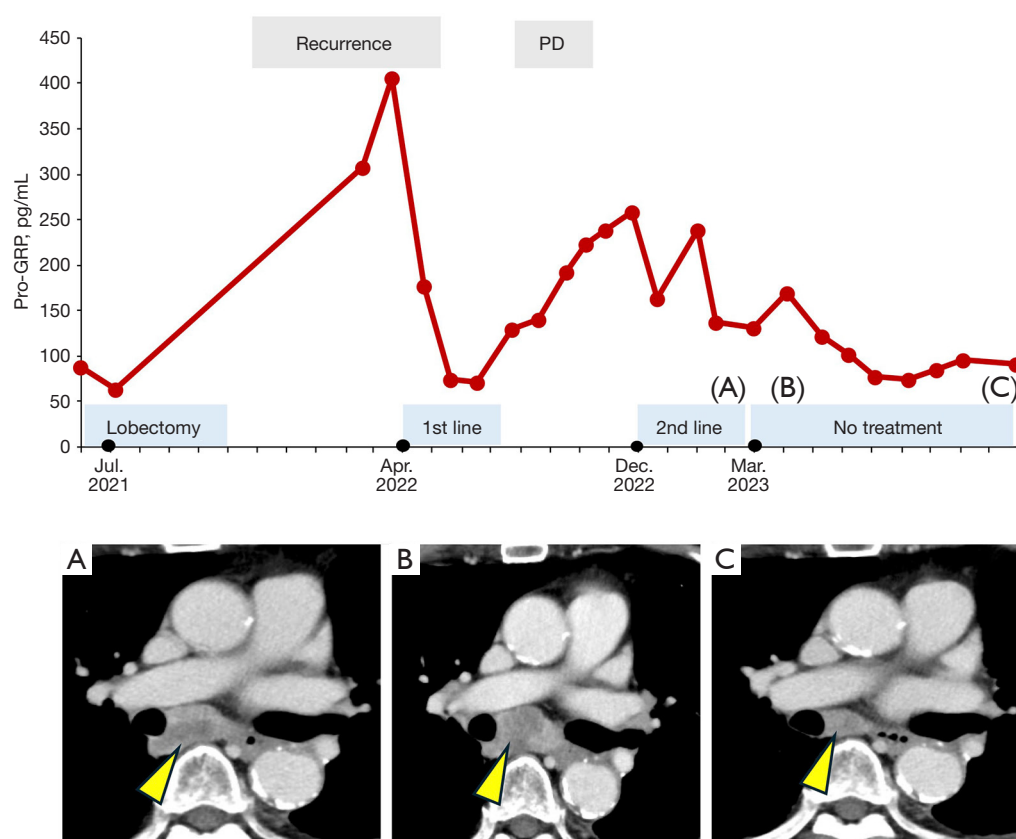


Figure 2 The clinical course is shown. Left lower lobectomy was performed in July 2021, first-line treatment began in April 2022, second-line treatment began in December 2022, and no treatment has been administered since March 2023. The CT images display the subcarinal lymph node (arrowhead) at three time points: (A) at the end of second-line treatment, (B) when it temporarily enlarged, and (C) when it subsequently maintained shrinkage. CT, computed tomography; PD, progressive disease.

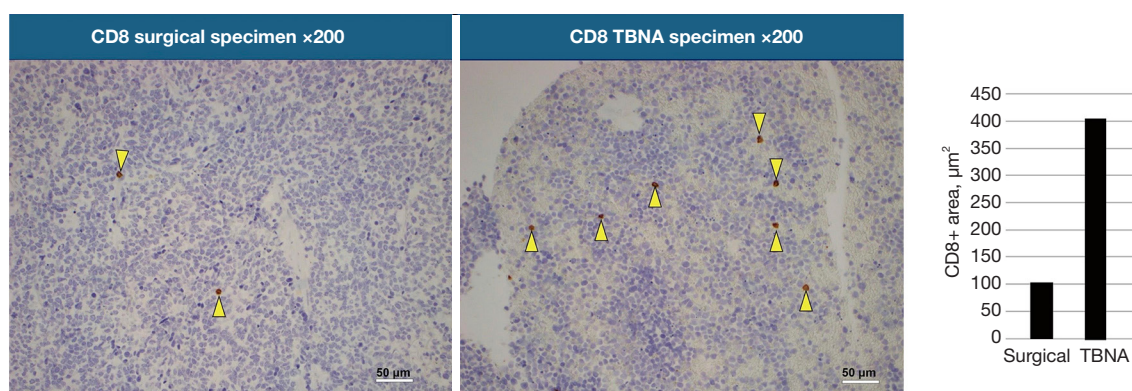


Figure 3 An increase in CD8-positive T cells (arrowheads) expression was observed in the TBNA specimens from the surgical ones (immunohistochemistry staining). CD8-positive T cells are quantified as shown in the graph. TBNA, transbronchial needle aspiration.

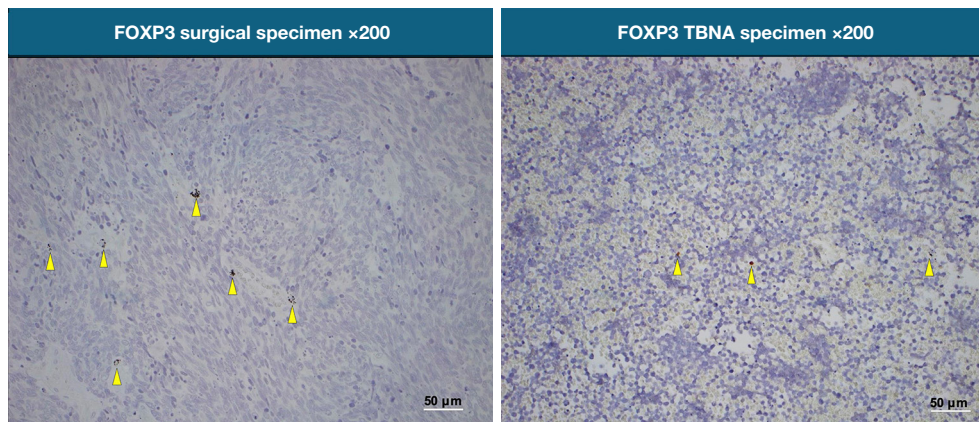


Figure 4 A decrease in regulatory T cells (arrowheads) expression was observed visually in the TBNA specimens from the surgical ones (immunohistochemistry staining). We cannot quantify them because of low expression. TBNA, transbronchial needle aspiration.

but it was difficult to quantify them due to low expression (Figure 4).

This suggests that the tumor microenvironment at recurrence had already undergone changes that may have prepared it for the subsequent immune modulation by chemotherapy. Previous reports have suggested an association between CD8-positive T cells and SR in lung cancer (7,8). In other previous reports, taxane-based anticancer drugs did not affect the viability of CD4-positive or CD8-positive T cells but induced apoptosis in regulatory T cells, selectively reducing their numbers and immunosuppressive function (9,10). In the current case, chemotherapy with nab-paclitaxel may have changed the tumor microenvironment, leading to an increase in CD8-positive T cells and a decrease in regulatory T cells, leading to SR.

Conclusions

Herein, we report a case of SR in a patient with SCLC. In previous reports on SR of cancer, there have been various discussions regarding the mechanism of action. We hypothesized that an increase in the number of CD8-positive cells at recurrence indicates that the tumor microenvironment has already been primed for immune modulation by chemotherapy, which enhances post-treatment immune activation, leading to SR. Further investigation of SR may offer insights into tumor immunity and future therapeutic strategies.

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Footnote

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Ethical Statement: The authors are accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All of the procedures performed in this case report were in accordance with the ethical standards of Juntendo University Hospital and with the Helsinki Declaration and its subsequent amendments. Written informed consent for publication of this case report and accompanying images was not obtained from the patient or the relatives after all possible attempts were made.

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